

1 Power Analysis for Longitudinal Clinical Trials with Run-In and
2 Common Close Observations

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5 Abstract

6 **Background.** Longitudinal clinical trials often incorporate pre-randomization
7 (run-in) and post-treatment (common close) observations to improve estimator efficiency.
8 Closed-form power and sample-size formulas for these augmented designs under linear
9 mixed-effects models with random slopes have remained scattered across the literature.
10 They are also only partially specified, so applied sample-size calculation becomes cum-
11 ber-some. **Methods.** We derive closed-form power and sample-size formulas for three nested
12 cases: (i) two run-in observations, (ii) an arbitrary number J_0 of run-in observations,
13 and (iii) J_0 run-in plus J_2 common close observations. All derivations proceed via the
14 Woodbury matrix identity applied to the GLS estimator under change-score parameteri-
15 zation. We express the formulas in terms of the residual variance σ^2 , the random-slope
16 variance σ_b^2 , and the observation spacing t , which enables direct computation of required
17 sample sizes. **Results.** We use parameters from Alzheimer’s disease neuroimaging
18 trials for a numerical example, where the signal-to-noise ratio $\sigma_b/\sigma \approx 1.7$ is favorable.
19 Two run-in observations reduce the required sample size by 62% (from 385 to 145 per
20 group), and adding two common close observations raises the reduction to 82%. The
21 magnitude of the gain depends on σ_b/σ . Across a representative range of slope variability,
22 the reduction from two run-in observations spans roughly 14% at low ratios to 76%
23 at high ratios. The closed-form expressions let us run this sensitivity analysis with
24 respect to σ_b/σ without re-running simulation. **Conclusions.** We report closed-form
25 expressions, derived via the Woodbury identity, for the variance of the treatment-effect
26 estimator under random-slopes longitudinal-trial designs with run-in and common close
27 observations. These results serve as a foundation for the AR(1) extension (Paper 2), the
28 optimal-allocation problem (Paper 3), and the GLS-versus-ANCOVA-versus-averaged
29 estimator comparison (Paper 4) in this compendium.

30 Contents

31	1 Introduction	2
32	2 Model	4
33	2.1 General formulation	4
34	2.2 Change-score parameterization	5
35	2.3 Covariance structure	6

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36	3 Variance of the Treatment Effect Estimator	6
37	3.1 GLS estimation and the Woodbury identity	6
38	3.2 Case 1: Standard design, no run-in ($J_0 = 0, J_1 = 2, J_2 = 0$)	7
39	3.3 Case 2: Two run-in observations ($J_0 = 2, J_1 = 1, J_2 = 0$)	8
40	3.4 General formula: J_0 run-in observations ($J_2 = 0$)	11
41	3.5 General formula with common close ($J_2 \geq 0$)	13
42	4 Power and Sample Size Formulas	14
43	5 Numerical Examples	15
44	5.1 Verification of closed-form results	15
45	5.2 Validation of the general routine	17
46	5.3 Sample size as a function of run-in observations	18
47	5.4 Sample size as a function of common close observations	18
48	5.5 Relative efficiency surface	18
49	5.6 Application: Alzheimer's disease imaging trial	18
50	6 Discussion	20
51	7 Conflict of interest	21
52	8 Funding	21
53	9 Data availability statement	21
54	10 Reproducibility	22
55	11 References	22

56 1 Introduction

57 The run-in period has served multiple purposes in clinical trial design, from identifying compli-
58 ant participants to washing out prior treatments¹. A distinct and methodologically important
59 use has since emerged in trials where the primary outcome is a rate of change. There, pre-
60 randomization observations estimate each participant's baseline trajectory, and incorporating
61 that estimate into the analysis model substantially increases statistical power.

62 ² established the theoretical foundation for this approach within the linear mixed effects
63 framework. Their key insight was straightforward: when between-subject variability in rates of
64 change is large relative to measurement error, pre-randomization observations carry valuable
65 information about each participant's underlying trajectory. That information reduces the
66 residual variance of the treatment effect estimator. Working with neuroimaging outcomes
67 in Alzheimer's disease trials, where the ratio of slope variability to measurement error is
68 particularly favorable, they demonstrated sample size reductions of 30–45%.

69 ³ extended this framework by proposing two-period linear mixed effects models that jointly
70 model run-in and post-randomization data, rather than treating run-in observations merely
71 as covariates. Their approach yields additional power gains of up to 15% and enables unequal

randomization ratios without power loss. This follows because run-in observations effectively contribute placebo-equivalent information for all participants.

A parallel literature has developed closed-form variance and sample size formulae for the random coefficient regression model without a run-in phase.⁴ derived the variance of the constrained longitudinal data analysis estimator and its relation to analysis of covariance, and⁵ established efficiency conditions under which the two approaches coincide.⁶ generalized these results to random coefficient regression models with monotone missing data and variable follow-up, providing variance, power, and sample size expressions illustrated with data from the Alzheimer's Disease Neuroimaging Initiative. These formulae apply to designs with $J_0 = 0$ run-in observations and an arbitrary post-randomization visit schedule, and serve as the baseline case that we extend here.

The CRAN reference implementation of the $J_0 = 0$ formulae is the `longpower` package⁷. It provides the routines `edland.linear.power()`, `diggle.linear.power()`, and `liu.liang.linear.power()` for slope-comparison trials with random intercepts and slopes⁸⁻¹⁰. A complementary implementation is the Stata package `slopepower`¹¹, which obtains power for slope-comparison trials by simulating from a fitted mixed model rather than by evaluating a closed form. It addresses the same design question, but it does not cover run-in or common close phases, and its simulation route does not expose the variance as an explicit function of the design counts. We show in Section 3.3 that the closed-form result derived below reduces to the Ard-Edland slope-variance formula implemented in `edland.linear.power()` at the boundary $J_0 = J_2 = 0$, and to the Diggle compound-symmetry formula when additionally $\sigma_b^2 = 0$. The generalization developed here therefore embeds the existing `longpower` formulae as the baseline case against which the efficiency gains from run-in and common close observations are measured.

Despite these advances, no general closed-form power formula has been derived that simultaneously accounts for an arbitrary number of run-in observations and a separate set of common close observations collected after treatment cessation. The common close phase, in which all participants are observed off-treatment following the randomized period, provides additional information about individual trajectories at no cost to the treatment comparison. Post-treatment observation periods appear in delayed-start and randomized withdrawal designs aimed at distinguishing symptomatic from disease-modifying effects¹². However, the contribution of the common close to statistical efficiency within the GLS framework has not been formally characterized. The delayed-start design¹³ is the closest relative. It also observes participants after the initial randomized period, but it does so to test whether an early-treatment group retains an advantage over a late-start group. That is, the post-randomization phase carries its own treatment contrast and its own estimand. The common close considered here serves the different purpose of sharpening the estimate of each participant's underlying slope, with all participants off treatment and no contrast attached to the phase itself. A formal comparison of the two estimands, and of the assumptions each requires about post-cessation trajectories, is beyond the scope of this paper. The common close construction's own assumption (a frozen, non-decaying treatment offset carried through the common close phase) is stated explicitly where the change score is defined below, in the 'General formula with

common close' subsection ('Carryover assumption' paragraph).

We present three results. First, we derive a simple closed-form variance formula for the treatment effect estimator when exactly two run-in observations precede randomization (Section 3.2). Second, we generalize this formula to an arbitrary number J_0 of run-in observations (Section 3.3). Third, we further extend to include J_2 common close observations collected after treatment cessation (Section 3.4). All derivations proceed from the generalized least squares estimator under the Woodbury matrix identity, yielding expressions in terms of the residual variance σ^2 , the random slope variance σ_b^2 , and the observation spacing t . Section 5 provides numerical examples using parameters estimated from Alzheimer's disease neuroimaging studies.

2 Model

2.1 General formulation

We consider a two-arm longitudinal clinical trial in which N participants are observed at equally spaced time points across three phases: a run-in period (pre-randomization), a treatment period (post-randomization), and an optional common close period (post-treatment). Let Y_{ij} denote the outcome for participant i at time point j . The observation schedule is:

- **Run-in:** $j = -J_0, -J_0 + 1, \dots, -1$ (all participants untreated)
- **Randomization visit:** $j = 0$ (baseline)
- **Treatment period:** $j = 1, 2, \dots, J_1$ (participants randomized to treatment or placebo)
- **Common close:** $j = J_1 + 1, J_1 + 2, \dots, J_1 + J_2$ (all participants off-treatment)

Throughout, the phase counts (J_0, J_1, J_2) correspond one-to-one to the arguments **r** (run-in), **k** (post-randomization), and **f** (common close) of the accompanying **runinpower** package functions used in the code chunks below.

We model the outcome as

$$Y_{ij} = \alpha + a_i + (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i)t_j + u_{ij} + \varepsilon_{ij} \quad (1)$$

where α is the overall intercept, $a_i \sim N(0, \sigma_a^2)$ is a random intercept, $b_i \sim N(0, \sigma_b^2)$ is a random slope, $u_{ij} \sim N(0, \sigma_u^2)$ are within-subject between-visit random effects independent across visits, and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ are independent between-scan measurement errors. This three-component within-subject variance structure follows² (their Eq. 17). Because u_{ij} and ε_{ij} are both independent across visits, under a single scan per visit they enter every formula below only through their sum. We therefore write $w_{ij} = u_{ij} + \varepsilon_{ij}$ for the total within-visit residual, with

$$\sigma^2 \equiv \text{Var}(w_{ij}) = \sigma_u^2 + \sigma_\varepsilon^2. \quad (2)$$

The two within-visit components are separately identifiable only when a visit carries replicate scans. For the single-scan designs treated here, only their sum σ^2 enters the variance of the

149 treatment-effect estimator. The phase indicator is

$$150 \quad h_j = \begin{cases} 0 & \text{if } j \leq 0 \quad (\text{run-in}) \\ 1 & \text{if } j > 0 \quad (\text{post-randomization}) \end{cases}$$

151 and the treatment group indicator is

$$152 \quad g_i = \begin{cases} 1 & i = 1, \dots, m \quad (\text{treatment}) \\ 0 & i = m + 1, \dots, N \quad (\text{placebo}) \end{cases}$$

153 The fixed effects are: δ , the rate of change during the run-in period (common to all partici-
154 pants); β , the rate of change during the post-randomization period for the placebo group; and
155 γ , the treatment effect on the rate of change. We assume equally spaced observations with
156 interval t , so $t_j = jt$.

157 For the common close phase ($j > J_1$), both treatment and placebo groups revert to a common
158 off-treatment slope. We model this by setting $g_i = 0$ for all participants during the common
159 close, so that the slope for both groups is $\beta + b_i$ in this phase. This is equivalent to replacing
160 the treatment indicator with $g_i \cdot I(1 \leq j \leq J_1)$ in model (1).

161 2.2 Change-score parameterization

162 Following², we work with change scores relative to the randomization visit:

$$163 \quad c_{ij} = Y_{ij} - Y_{i0} \tag{3}$$

164 This eliminates the intercept α and the random intercept a_i . We note that the change-score
165 approach is algebraically equivalent to constrained longitudinal data analysis (cLDA) under
166 compound symmetry. There, the baseline is included in the response vector with a common-
167 mean constraint across treatment groups^{14,15}. The variance formulas derived below therefore
168 apply to both change-score and cLDA analyses of run-in designs. Because the change-score
169 eliminates a_i , the resulting formulas depend on only two variance components (σ^2, σ_b^2) rather
170 than the three ($\sigma^2, \sigma_a^2, \sigma_b^2$) required by the raw-outcome model.¹⁶ show that ignoring the
171 intercept-slope correlation in the three-component model can produce anticonservative sample
172 sizes. The change-score approach avoids this issue by construction, at the cost of discarding
173 information about σ_a^2 .

174 The change-score formulation yields

$$175 \quad c_{ij} = (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i) t_j + (w_{ij} - w_{i0}) \tag{4}$$

176 for $j \neq 0$. The change-score vector for participant i has dimension $J = J_0 + J_1 + J_2$, encom-
177 passing all three phases.

2.3 Covariance structure

The marginal covariance of the change-score vector is

$$\Sigma = R + ZGZ' \quad (5)$$

where $G = \sigma_b^2$ (scalar, since we have a single random slope), $Z = t \cdot \mathbf{1}_J$ is the $J \times 1$ random effects design vector, and R is the $J \times J$ residual covariance of the differenced within-visit residuals $(w_{ij} - w_{i0})$.

The residual covariance has the structure

$$R_{jl} = \text{Cov}(w_{ij} - w_{i0}, w_{il} - w_{i0}) = \begin{cases} 2\sigma^2 & \text{if } j = l \\ \sigma^2 & \text{if } j \neq l \end{cases} \quad (6)$$

This follows because $w_{ij} = u_{ij} + \varepsilon_{ij}$ are independent across visits with common variance $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2$. The shared baseline residual w_{i0} induces positive correlation among all change scores. Thus $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$. Both within-visit components enter R identically because u_{i0} and ε_{i0} are differenced out of every change score in the same way. This is why only their sum σ^2 is identified from single-scan designs.

3 Variance of the Treatment Effect Estimator

3.1 GLS estimation and the Woodbury identity

The generalized least squares estimator of the fixed effects is

$$\hat{\beta} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c} \quad (7)$$

with $\text{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$, where X is the fixed effects design matrix and $\mathbf{c}$ is the stacked change-score vector. Our target is $\text{Var}(\hat{\gamma})$, the variance of the treatment effect estimator, which is the diagonal element of $(X'\Sigma^{-1}X)^{-1}$ corresponding to γ .

The key computational tool is the Woodbury matrix identity. Since $\Sigma = R + ZGZ'$ with scalar $G = \sigma_b^2$:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z(G^{-1} + Z'R^{-1}Z)^{-1}Z'R^{-1} \quad (8)$$

This avoids direct inversion of the $J \times J$ matrix Σ and enables closed-form results because R^{-1} has a known simple structure.

From $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$, the Sherman-Morrison formula gives

$$R^{-1} = \frac{1}{\sigma^2} \left(\mathbf{I}_J - \frac{1}{J+1} \mathbf{1}_J\mathbf{1}_J' \right) \quad (9)$$

205 With $Z = t \cdot \mathbf{1}_J$, we obtain

$$206 \quad Z' R^{-1} Z = \frac{t^2}{\sigma^2} \left(J - \frac{J^2}{J+1} \right) = \frac{J t^2}{\sigma^2(J+1)} \quad (10)$$

207 and

$$208 \quad (G^{-1} + Z' R^{-1} Z)^{-1} = \frac{\sigma^2 \sigma_b^2 (J+1)}{\sigma^2(J+1) + J t^2 \sigma_b^2} \quad (11)$$

209 Let $a = \sigma^2$ and $b = \sigma_b^2$ for conciseness. Define

$$210 \quad \eta_J = \frac{a(J+1) + J t^2 b}{a(J+1)} = 1 + \frac{J t^2 b}{a(J+1)} \quad (12)$$

211 so that $(G^{-1} + Z' R^{-1} Z)^{-1} = b/\eta_J$.

212 3.1.1 Per-participant information, compact form

213 Let X_i , Z_i , and R_i denote the per-participant fixed effects design, random effects design, and
214 residual covariance, and define the scalar

$$215 \quad \mathcal{J}_i = Z_i' R_i^{-1} Z_i + G^{-1} \quad (13)$$

216 By (8), the per-participant contribution to the GLS information matrix is

$$217 \quad X_i' \Sigma_i^{-1} X_i = X_i' R_i^{-1} X_i - \frac{(X_i' R_i^{-1} Z_i)(Z_i' R_i^{-1} X_i)}{\mathcal{J}_i} \quad (14)$$

218 and the total information is $X' \Sigma^{-1} X = \sum_i X_i' \Sigma_i^{-1} X_i$, with $\text{Var}(\hat{\beta}) = (X' \Sigma^{-1} X)^{-1}$. Two
219 features of this form are used later. First, under a single random slope, $\mathcal{J}_i$ is a scalar, so
220 (14) is a rank-one correction of the OLS-style information $X_i' R_i^{-1} X_i$. Second, because per-
221 participant contributions add, heterogeneous designs (varying J_0 , J_2 , or dropout patterns)
222 are accommodated by summing (14) over the relevant strata.

223 3.2 Case 1: Standard design, no run-in ($J_0 = 0$, $J_1 = 2$, $J_2 = 0$)

224 This case reproduces the result in², Section 3.1.1, and is the $J_0 = 0$ baseline of the random
225 coefficient regression power formulae of⁴ and⁶. It serves as verification for our framework.

226 With $J_0 = 0$ and $J_1 = 2$, there are $J = 2$ change scores $(c_{i,1}, c_{i,2})$ at post-randomization times
227 $(t, 2t)$. Since there is no run-in, the model has two fixed effects: the placebo slope β and the
228 treatment effect γ .

229 The design matrices are:

$$230 \quad X_i^{(\text{trt})} = t \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}, \quad X_i^{(\text{plc})} = t \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$$

231 The residual covariance is $R = a \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with $R^{-1} = \frac{1}{3a} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, and the random effects
 232 design is $Z = t(1, 2)'$.

233 Applying the Woodbury identity with $J = 2$:

$$234 \quad Z'R^{-1}Z = \frac{t^2}{3a}(1, 2) \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{t^2}{3a}(1, 2)(0, 3)' = \frac{6t^2}{3a} = \frac{2t^2}{a} \quad (15)$$

$$235 \quad (G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a + 2bt^2} \quad (16)$$

237 After summing over one treatment and one placebo participant, the information matrix is

$$238 \quad X'\Sigma^{-1}X = \frac{2t^2}{a + 2bt^2} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad (17)$$

239 and its inverse is

$$240 \quad \text{Var}(\hat{\beta}) = \frac{a + 2bt^2}{2t^2} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \quad (18)$$

241 The variance of the treatment effect estimator (per participant pair) is the (2, 2) element:

$$242 \quad \boxed{\text{Var}(\hat{\gamma})|_{J_0=0, J_1=2} = \frac{a + 2bt^2}{t^2} = \frac{\sigma^2 + 2\sigma_b^2 t^2}{t^2}} \quad (19)$$

243 This agrees with Frost et al. (2008), Section 3.1.1. For m participants per group: $\text{Var}(\hat{\gamma}) =$
 244 $(\sigma^2 + 2\sigma_b^2 t^2)/(mt^2)$. This serves as the baseline against which the run-in designs are compared.

245 **3.3 Case 2: Two run-in observations ($J_0 = 2, J_1 = 1, J_2 = 0$)**

246 We now have $J = 3$ change scores ($c_{i,-2}, c_{i,-1}, c_{i,1}$) at times $(-2t, -t, t)$, the baseline visit
 247 at time 0 having been differenced away. The fixed effects design matrix separates the run-in
 248 slope δ from the post-randomization slopes:

$$249 \quad X_i^{(\text{trt})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & t \end{pmatrix}, \quad X_i^{(\text{plc})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & 0 \end{pmatrix}$$

250 Since the random slope applies uniformly to all time points, $Z = t(-2, -1, 1)'$ for the random
 251 effect b_i . We note that the change scores at times $-2t$ and $-t$ relative to baseline at 0 yield
 252 $c_{i,-2} = Y_{i,-2} - Y_{i,0}$ and $c_{i,-1} = Y_{i,-1} - Y_{i,0}$.

253 More carefully, under model (1):

$$\begin{aligned}
254 \quad c_{i,-2} &= -2(\delta + b_i)t + (w_{i,-2} - w_{i,0}) \\
255 \quad c_{i,-1} &= -(\delta + b_i)t + (w_{i,-1} - w_{i,0}) \\
256 \quad c_{i,1} &= (\beta + \gamma g_i + b_i)t + (w_{i,1} - w_{i,0})
\end{aligned} \tag{20}$$

257 Thus the design matrices for the fixed effects $(\delta, \beta, \gamma)'$ are:

$$258 \quad X_i^{(\text{trt})} = t \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad X_i^{(\text{plc})} = t \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

259 and the random effects design is $Z_i = t(-2, -1, 1)'$.

260 The residual covariance is

$$261 \quad R = a \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = a(\mathbf{I}_3 + \mathbf{1}_3 \mathbf{1}_3') \tag{21}$$

262 with inverse

$$263 \quad R^{-1} = \frac{1}{a} \left(\mathbf{I}_3 - \frac{1}{4} \mathbf{1}_3 \mathbf{1}_3' \right) = \frac{1}{4a} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \tag{22}$$

264 Now we compute the Woodbury components. With $Z = t(-2, -1, 1)'$:

$$265 \quad Z' R^{-1} Z = \frac{t^2}{4a} (3 \cdot 4 + 3 \cdot 1 + 3 \cdot 1 - 2(-1)(-2)(-1) - 2(-1)(-2)(1) - 2(-1)(-1)(1)) \tag{23}$$

$$266 \quad \text{Computing directly: } Z' R^{-1} Z = \frac{t^2}{4a} (-2, -1, 1) \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} (-2, -1, 1)'$$

$$267 \quad R^{-1} Z = \frac{t}{4a} \begin{pmatrix} 3(-2) + (-1)(-1) + (-1)(1) \\ (-1)(-2) + 3(-1) + (-1)(1) \\ (-1)(-2) + (-1)(-1) + 3(1) \end{pmatrix} = \frac{t}{4a} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix}$$

$$268 \quad \text{so } Z' R^{-1} Z = \frac{t^2}{4a} [(-2)(-6) + (-1)(-2) + (1)(6)] = \frac{t^2}{4a} (12 + 2 + 6) = \frac{20t^2}{4a} = \frac{5t^2}{a}.$$

$$269 \quad \text{Alternatively, we note } Z = t \cdot \mathbf{d} \text{ where } \mathbf{d} = (-2, -1, 1)', \text{ so } Z' R^{-1} Z = \frac{t^2}{a} (\mathbf{d}' \mathbf{d} - \frac{1}{4} (\mathbf{1}' \mathbf{d})^2) =$$

$$270 \quad \frac{t^2}{a} (6 - \frac{4}{4}) = \frac{5t^2}{a}.$$

271 Thus

$$272 \quad H \equiv (G^{-1} + Z' R^{-1} Z)^{-1} = \frac{ab}{a + 5bt^2} \tag{24}$$

273 For the information matrix, we compute $X'R^{-1}X$ summed over treatment and placebo groups.

274 For a single treatment participant:

$$275 \quad X_i^{(\text{trt})'} R^{-1} X_i^{(\text{trt})} = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

276 Computing the product:

$$277 \quad X_i^{(\text{trt})'} R^{-1} X_i^{(\text{trt})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix} \quad (25)$$

278 For placebo (whose γ column is zero):

$$279 \quad X_i^{(\text{plc})'} R^{-1} X_i^{(\text{plc})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (26)$$

280 Summing (one participant per group for the derivation, then multiplying by m):

$$281 \quad X'R^{-1}X = \frac{t^2}{4a} \begin{pmatrix} 22 & 6 & 3 \\ 6 & 6 & 3 \\ 3 & 3 & 3 \end{pmatrix} \quad (27)$$

282 For the Woodbury correction, we need $X'R^{-1}Z$. For treatment:

$$283 \quad X_i^{(\text{trt})'} R^{-1} Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix}$$

284 For placebo:

$$285 \quad X_i^{(\text{plc})'} R^{-1} Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix}$$

286 The Woodbury correction requires the per-subject outer products (not the outer product of

the sum):

$$\begin{aligned}
W &= \sum_g X_i^{(g)'} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)} \\
&= \frac{bt^4}{16a(a+5bt^2)} \left(\begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix} (14, 6, 6) + \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix} (14, 6, 0) \right) \\
&= \frac{bt^4}{16a(a+5bt^2)} \begin{pmatrix} 392 & 168 & 84 \\ 168 & 72 & 36 \\ 84 & 36 & 36 \end{pmatrix}
\end{aligned} \tag{28}$$

Therefore

$$X' \Sigma^{-1} X = X' R^{-1} X - W \tag{29}$$

To find $\text{Var}(\hat{\gamma})$, we need the $(3, 3)$ element of $(X' \Sigma^{-1} X)^{-1}$. We compute this numerically in the R code below and verify against the closed-form result.

Carrying through the algebra, define $D = a + 5bt^2 = \sigma^2 + 5\sigma_b^2 t^2$ and write $X' \Sigma^{-1} X = \frac{t^2}{16aD} A$. The matrix A has $(3, 3)$ minor $1536aD$ and determinant $9216aD(a+2bt^2)$. Hence the $(3, 3)$ cofactor of $X' \Sigma^{-1} X$ is $1536aD \cdot t^4 / (16aD)^2 = 6t^4 / (aD)$, and its determinant is $9216aD(a+2bt^2) \cdot t^6 / (16aD)^3$. The ratio gives:

$$\boxed{\text{Var}(\hat{\gamma})|_{J_0=2, J_1=1} = \frac{8\sigma^2(\sigma^2 + 5\sigma_b^2 t^2)}{3t^2(\sigma^2 + 2\sigma_b^2 t^2)}} \tag{30}$$

Unlike the Frost formula (19), the run-in variance depends on σ^2 both in the numerator and denominator, reflecting the additional information extracted from pre-randomization observations.

3.4 General formula: J_0 run-in observations ($J_2 = 0$)

We now derive the variance for arbitrary J_0 run-in observations with J_1 post-randomization observations and no common close ($J_2 = 0$). The total number of change scores is $J = J_0 + J_1$.

The change scores are

$$\begin{aligned}
c_{i,-s} &= -s(\delta + b_i)t + (w_{i,-s} - w_{i0}) & s &= 1, \dots, J_0 \\
c_{i,l} &= l(\beta + \gamma g_i + b_i)t + (w_{i,l} - w_{i0}) & l &= 1, \dots, J_1
\end{aligned} \tag{31}$$

The fixed effects design matrix has columns for (δ, β, γ) . Let $\mathbf{d}_{J_0} = (-J_0, -J_0 + 1, \dots, -1)' \cdot t$ be the run-in time vector and $\mathbf{d}_{J_1} = (1, 2, \dots, J_1)' \cdot t$ be the post-randomization time vector. Then:

$$X_i^{(\text{trt})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{d}_{J_1} \end{pmatrix}, \quad X_i^{(\text{plc})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{0} \end{pmatrix}$$

313 The random effects design vector is $Z_i = (\mathbf{d}'_{J_0}, \mathbf{d}'_{J_1})'$, i.e., the full vector of time distances
 314 from baseline.

315 The residual covariance remains $R = a(\mathbf{I}_J + \mathbf{1}_J \mathbf{1}'_J)$ with inverse given by (9).

316 **3.4.1 Computing $X'\Sigma^{-1}X$**

317 We proceed through the Woodbury identity. Define $S_{J_0} = \sum_{s=1}^{J_0} s^2 = J_0(J_0 + 1)(2J_0 + 1)/6$
 318 and $S_{J_1} = \sum_{l=1}^{J_1} l^2 = J_1(J_1 + 1)(2J_1 + 1)/6$.

319 The key intermediate quantities are:

320 $Z'R^{-1}Z$: With $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1)'$:

$$321 \quad Z'R^{-1}Z = \frac{t^2}{a} \left[(S_{J_0} + S_{J_1}) - \frac{(\sum d_j)^2}{J+1} \right] \quad (32)$$

322 where $\sum d_j = -J_0(J_0 + 1)/2 + J_1(J_1 + 1)/2$.

323 $(G^{-1} + Z'R^{-1}Z)^{-1}$: With $Z'R^{-1}Z$ given by (32),

$$324 \quad H = (G^{-1} + Z'R^{-1}Z)^{-1} = \frac{b}{1 + b Z'R^{-1}Z},$$

325 the general analogue of b/η_J in (12).

326 $X'R^{-1}X$: This 3×3 matrix has elements computed from the block structure of X . Letting
 327 $D_{J_0} = \sum_{s=1}^{J_0} s = J_0(J_0 + 1)/2$ and $D_{J_1} = \sum_{l=1}^{J_1} l = J_1(J_1 + 1)/2$, and using $R^{-1} = \frac{1}{a}(\mathbf{I}_J -$
 328 $\frac{1}{J+1}\mathbf{1}\mathbf{1}')$, the information matrix summed over one treatment and one placebo participant is:

$$329 \quad X'R^{-1}X = \frac{t^2}{a} \begin{pmatrix} 2\left(S_{J_0} - \frac{D_{J_0}^2}{J+1}\right) & \frac{2D_{J_0}D_{J_1}}{J+1} & \frac{D_{J_0}D_{J_1}}{J+1} \\ \frac{2D_{J_0}D_{J_1}}{J+1} & 2\left(S_{J_1} - \frac{D_{J_1}^2}{J+1}\right) & S_{J_1} - \frac{D_{J_1}^2}{J+1} \\ \frac{D_{J_0}D_{J_1}}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} \end{pmatrix} \quad (33)$$

330 The factor of 2 appears on entries involving both groups, while γ -related entries lack this factor
 331 since the γ column is zero for the placebo group. The positive sign on the (δ, β) cross-term
 332 arises because the run-in and post-randomization time sums have opposite signs ($\sum d_{J_0} < 0$,
 333 $\sum d_{J_1} > 0$).

334 **Woodbury correction:** The correction is the sum of per-subject rank-one outer products:

$$335 \quad W = \sum_{g \in \{\text{trt}, \text{plc}\}} X_i^{(g)'} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)}$$

336 which is generally rank 2 (not rank 1), so $X'\Sigma^{-1}X = X'R^{-1}X - W$.

3.4.2 Numerical implementation

The variance $\text{Var}(\hat{\gamma})$ for general J_0 and J_1 does not admit a simple closed-form expression. The formula depends on σ^2 and σ_b^2 in a ratio involving S_{J_0} , S_{J_1} , D_{J_0} , D_{J_1} , and J . Moreover, the interaction between the Woodbury correction and the information matrix produces algebraically complex expressions.

We therefore implement the computation via the matrix-based Woodbury identity in R code, which evaluates the exact variance for any configuration (J_0, J_1, J_2) . The closed-form expressions for $J_0 = 0$ (19) and $J_0 = 2$ (30) serve as verification checkpoints.

3.4.3 Remark: recovery of the Edland and Diggle formulae

At $J_0 = 0$, the δ column of X is empty and the information matrix reduces to a 2×2 block on (β, γ) . Carrying the Woodbury algebra through, the Sherman-Morrison cancellations collapse the Woodbury correction against the $(2, 1; 1, 1)$ structure of $X'R^{-1}X$. The variance of the treatment effect estimator for n participants per arm then becomes

$$\text{Var}(\hat{\gamma})|_{J_0=0, J_2=0} = \frac{2}{n} \left[\sigma_b^2 + \frac{\sigma^2}{S_{xx}} \right], \quad S_{xx} = \sum_{j=0}^{J_1} (t_j - \bar{t})^2, \quad (34)$$

where the baseline visit at $t_0 = 0$ enters through the change-score transformation. Equation (34) is the slope-comparison formula of⁸ implemented in `longpower::edland.linear.power()`. Setting additionally $\sigma_b^2 = 0$ reduces Σ to compound symmetry and recovers the formula of⁹ used by `longpower::diggle.linear.power()`. The Frost formula (19) is in turn the $J_1 = 2$ specialization of (34), since $S_{xx} = 2t^2$ at that setting. The general- (J_0, J_1, J_2) result derived in the next subsection therefore embeds both `longpower` closed-form routines as strict boundary cases. The efficiency gains quantified in Section 5 accrue entirely from the information contributed by $J_0 > 0$ or $J_2 > 0$.

3.5 General formula with common close ($J_2 \geq 0$)

We now extend to include J_2 common close observations after the treatment period. The total dimension is $J = J_0 + J_1 + J_2$.

During the common close phase (time points $J_1 + 1, \dots, J_1 + J_2$), all participants are off-treatment. The change scores are:

$$c_{i, J_1+s} = [\beta(J_1 + s) + \gamma g_i \cdot J_1 + b_i(J_1 + s)] t + (w_{i, J_1+s} - w_{i0}) \quad s = 1, \dots, J_2 \quad (35)$$

The treatment effect γ enters the common close change scores only through the accumulated effect during the treatment period ($\gamma g_i \cdot J_1 \cdot t$), not through the slope during the common close. This means the γ column in the design matrix has the value $J_1 t$ for all common close observations in the treatment group.

Carryover assumption. This encodes an explicit modeling choice. The between-group

offset acquired during treatment is held constant (frozen) through the common close phase, neither growing nor decaying once treatment stops. Equivalently, the common close slope is assumed equal between arms, so the two groups' trajectories run parallel after treatment ends, and the gap opened during the treatment period is carried forward unchanged. This is the same assumption distinguished from the delayed-start estimand in the Introduction's discussion of¹³ above. It is not derived from a biological mechanism and should be treated as a design simplification rather than an empirical finding. If post-cessation trajectories converge (loss of treatment benefit) or diverge (continued divergence after cessation), the variance and power expressions below remain correct algebraically. However, the estimand they target (the end-of-common-close group difference under a frozen offset) would no longer match the quantity of clinical interest.

The fixed effects design matrix becomes:

$$X_i^{(\text{trt})} = t \begin{pmatrix} -J_0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ -1 & 0 & 0 \\ 0 & 1 & 1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 & J_1 \\ 0 & J_1 + 1 & J_1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 + J_2 & J_1 \end{pmatrix}$$

The β column contains the post-randomization time for the placebo slope: $0, \dots, 0, 1, \dots, J_1, J_1 + 1, \dots, J_1 + J_2$. The γ column is $0, \dots, 0, 1, \dots, J_1, J_1, \dots, J_1$ for treatment participants and all zeros for placebo.

The random effects design remains $Z_i = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1, J_1 + 1, \dots, J_1 + J_2)'$.

The residual covariance is still $R = a(\mathbf{I}_J + \mathbf{1}_J \mathbf{1}_J')$ with $J = J_0 + J_1 + J_2$.

The Woodbury computation proceeds identically, with the only changes being the dimension J , the structure of the X matrix (specifically the γ column), and the elements of Z .

We implement the general computation in R below.

4 Power and Sample Size Formulas

For a two-sided test at significance level α with power $1 - \beta_{\text{pow}}$, the required number of participants per group is

$$m = \frac{(z_{\alpha/2} + z_{1-\beta_{\text{pow}}})^2 \cdot \text{Var}_1(\hat{\gamma})}{\Delta^2} \quad (36)$$

where $\text{Var}_1(\hat{\gamma})$ denotes the per-participant-pair variance from the formulas derived above, and Δ is the minimum clinically important treatment effect on the rate of change.

Equivalently, the power for a given sample size is

$$1 - \beta_{\text{pow}} = \Phi \left(\frac{\Delta \sqrt{m}}{\sqrt{\text{Var}_1(\hat{\gamma})}} - z_{\alpha/2} \right) \quad (37)$$

The relative efficiency of a design with (J_0, J_1, J_2) observations compared to a standard design with $(0, J_1, 0)$ is

$$\text{RE}(J_0, J_1, J_2) = \frac{\text{Var}_1(\hat{\gamma})|_{0, J_1, 0}}{\text{Var}_1(\hat{\gamma})|_{J_0, J_1, J_2}} \quad (38)$$

Values greater than 1 indicate that the run-in/common close design requires fewer participants.

A practical caveat is in order here. Adding a single run-in observation ($J_0 = 1$) introduces the run-in slope parameter δ into the model, increasing the fixed-effect dimension from 2 to 3. This additional parameter absorbs part of the information contributed by the run-in observation. Numerical evaluation over a wide grid of configurations ($J_1 = 2, \dots, 6$; σ_b^2/σ^2 from 10^{-4} to 10^3 ; t from 0.25 to 4) shows that $\text{Var}(\hat{\gamma})|_{J_0=1}$ never exceeds $\text{Var}(\hat{\gamma})|_{J_0=0}$. That said, the improvement can be negligible: at some parameter values the information from the single run-in observation is almost exactly offset by the cost of estimating δ . A single run-in observation is therefore never harmful to the precision of $\hat{\gamma}$, although the gain it provides should be quantified with the formulas above before committing to a longer trial. This is an instance of the general design principle set out by¹⁷: the value of an additional measurement occasion in a longitudinal trial depends jointly on the variance components and on the parameters the enlarged model must estimate. The number and placement of visits should accordingly be chosen by direct evaluation of the resulting precision rather than by a fixed rule.

5 Numerical Examples

5.1 Verification of closed-form results

We begin with a check that the closed-form expressions agree with the matrix computation. The Frost et al. (2008) result corresponds to the standard design with $J_0 = 0$, $J_1 = 2$ (two post-baseline visits at times t and $2t$), which under the two-parameter model yields $\text{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$. The $J_0 = 2$, $J_1 = 1$ case uses the three-parameter model.

```
a <- 1.0
b <- 0.5
tt <- 1.0

v_frost_matrix <- var_gamma_matrix(0, 2, 0, a, b, tt)
v_frost_closed <- var_gamma_frost(a, b, tt)

v_r2_matrix <- var_gamma_matrix(2, 1, 0, a, b, tt)
v_r2_closed <- var_gamma_r2(a, b, tt)
```

```

cat('Frost (r=0, k=2) case:\n')

424 ## Frost (r=0, k=2) case:
cat('  Matrix:', round(v_frost_matrix, 6), '\n')

425 ##   Matrix: 2
cat('  Closed:', round(v_frost_closed, 6), '\n')

426 ##   Closed: 2
cat('  Match:',
    all.equal(v_frost_matrix, v_frost_closed), '\n\n')

427 ##   Match: TRUE
cat('r=2, k=1 case:\n')

428 ## r=2, k=1 case:
cat('  Matrix:', round(v_r2_matrix, 6), '\n')

429 ##   Matrix: 4.666667
cat('  Closed:', round(v_r2_closed, 6), '\n')

430 ##   Closed: 4.666667
cat('  Match:',
    all.equal(v_r2_matrix, v_r2_closed), '\n\n')

431 ##   Match: TRUE
cat('Variance comparison across designs:\n')

432 ## Variance comparison across designs:
for (rr in 0:4) {
  v <- var_gamma_matrix(rr, 2, 0, a, b, tt)
  cat(sprintf('  r=%d, k=2: Var = %.4f\n', rr, v))
}

433 ##   r=0, k=2: Var = 2.0000
434 ##   r=1, k=2: Var = 2.0000
435 ##   r=2, k=2: Var = 1.7910
436 ##   r=3, k=2: Var = 1.5000
437 ##   r=4, k=2: Var = 1.2651

```


5.2 Validation of the general routine

The closed-form checkpoints above both have $J_2 = 0$, so they do not exercise the common-close path. We therefore validate the general routine in two ways. First, we compare the Woodbury computation with a direct GLS evaluation, in which $\Sigma = R + \sigma_b^2 Z Z'$ is assembled and inverted without the Woodbury identity, over a grid of designs that includes configurations with $J_2 > 0$. Second, we run a seeded Monte Carlo benchmark at $(J_0, J_1, J_2) = (2, 2, 2)$. Change-score vectors are simulated from the generative model, $\hat{\gamma}$ is computed by GLS in each replicate, and the empirical variance is compared with the analytic value.

```
var_gamma_direct <- function(r, k, f, sigma2, sigma_b2,
                             t_interval) {
  p <- r + k + f
  R <- sigma2 * (diag(p) + matrix(1, p, p))
  des <- build_design(r, k, f, t_interval)
  Sigma <- R + sigma_b2 * des$Z %*% t(des$Z)
  Si <- solve(Sigma)
  info <- t(des$X_trt) %*% Si %*% des$X_trt +
    t(des$X_plc) %*% Si %*% des$X_plc
  solve(info)[des$ncol_X, des$ncol_X]
}
```

```
grid <- expand.grid(r = 0:3, k = 1:3, f = 0:3,
                  sigma_b2 = c(0.1, 1, 5),
                  t_int = c(0.5, 1, 2))
grid <- grid[grid$r + grid$k + grid$f >= 3, ]
diffs <- mapply(
  function(r, k, f, sb2, tt) {
    abs(var_gamma_matrix(r, k, f, 1.0, sb2, tt) -
      var_gamma_direct(r, k, f, 1.0, sb2, tt))
  },
  grid$r, grid$k, grid$f, grid$sigma_b2, grid$t_int
)
```

```
cat('Direct GLS vs Woodbury routine over', nrow(grid),
    'configurations (including f > 0):\n')
```

```
## Direct GLS vs Woodbury routine over 396 configurations (including f > 0):
```

```
cat(' Maximum absolute difference:',
    format(max(diffs), digits = 3), '\n\n')
```

```
## Maximum absolute difference: 3.38e-14
```

```
set.seed(20260418)
r_v <- 2; k_v <- 2; f_v <- 2
s2_v <- 1; sb2_v <- 1; t_v <- 1
```

```

m_v <- 200; n_sim <- 2000
des <- build_design(r_v, k_v, f_v, t_v)
p_v <- r_v + k_v + f_v
Sigma <- s2_v * (diag(p_v) + matrix(1, p_v, p_v)) +
  sb2_v * des$Z %*% t(des$Z)
Si <- solve(Sigma)
cR <- chol(Sigma)
info <- m_v * (t(des$X_trt) %*% Si %*% des$X_trt +
  t(des$X_plc) %*% Si %*% des$X_plc)
theta <- c(0.2, -0.5, 0.3)
gamma_hat <- numeric(n_sim)
for (s in seq_len(n_sim)) {
  xty <- 0
  for (X in list(des$X_trt, des$X_plc)) {
    mu <- as.numeric(X %*% theta)
    Y <- matrix(rep(mu, m_v), m_v, p_v, byrow = TRUE) +
      matrix(rnorm(m_v * p_v), m_v, p_v) %*% cR
    xty <- xty + t(X) %*% Si %*% colSums(Y)
  }
  gamma_hat[s] <- solve(info, xty)[3]
}
cat('Monte Carlo at (J0, J1, J2) = (2, 2, 2), m =', m_v,
  'per group,', n_sim, 'replicates:\n')

```

448 ## Monte Carlo at (J0, J1, J2) = (2, 2, 2), m = 200 per group, 2000 replicates:

```

cat(' Empirical m * Var(gamma_hat):',
  round(m_v * var(gamma_hat), 4), '\n')

```

449 ## Empirical m * Var(gamma_hat): 1.1039

```

cat(' Theoretical (Woodbury):      ',
  round(var_gamma_matrix(r_v, k_v, f_v, s2_v, sb2_v,
    t_v), 4), '\n')

```

450 ## Theoretical (Woodbury): 1.1344

451 5.3 Sample size as a function of run-in observations

452 5.4 Sample size as a function of common close observations

453 5.5 Relative efficiency surface

454 5.6 Application: Alzheimer's disease imaging trial

455 We use variance parameters estimated from the MIRIAD dataset²: $\sigma_b^2 = 0.9745$ (between-
 456 subject slope variance) and a combined within-visit residual $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2 = 0.3338 + 0.0025 =$

Table 1: Required sample size per group ($J_1 = 2$, $J_2 = 0$, $\Delta = 0.5$, 90% power, two-sided $\alpha = 0.05$) as a function of the number of run-in observations J_0 , for three levels of between-subject slope variability.

J_0	Low	Medium	High
0	51	127	463
1	44	119	214
2	44	91	113
3	43	70	76
4	42	56	59
5	40	48	49
6	38	42	43

Table 2: Required sample size per group ($J_0 = 2$, $J_1 = 2$, $\Delta = 0.5$, 90% power) as a function of common close observations J_2 .

J_2	Low	Medium	High
0	44	91	113
1	34	69	80
2	29	48	52
3	25	35	37
4	22	27	28

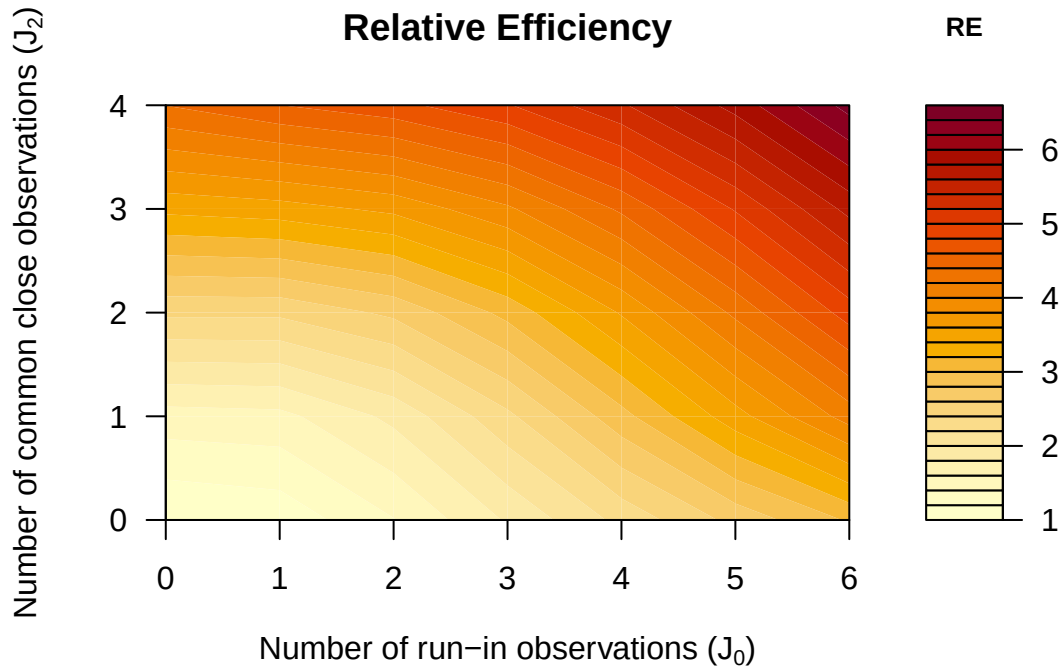


Figure 1: (#fig:fig_efficiency)Relative efficiency of run-in and common close designs compared to the standard design ($J_0 = 0$, $J_2 = 0$) with the same number of post-randomization observations ($J_1 = 2$). Parameters: $\sigma^2 = 1$, $\sigma_b^2 = 1$, $t = 1$.

Table 3: Sample size per group for AD imaging trial ($J_1 = 2$ post-randomization visits, $\Delta = 0.25$, 90% power). All $J_1 = 2$ configurations ordered by sample size; the (0, 2, 0) row is the standard-design baseline.

J_0	J_1	J_2	m per group	Total visits
4	2	2	49	9
3	2	2	58	8
4	2	1	60	8
2	2	2	69	7
3	2	1	77	7
0	2	2	77	5
1	2	2	77	6
4	2	0	78	7
3	2	0	100	6
2	2	1	104	6
1	2	1	138	5
2	2	0	145	5
0	2	1	150	4
1	2	0	247	4
0	2	0	385	3

0.3363, formed per Eq. (2) from the within-subject between-visit component ($\sigma_u^2 = 0.3338$) and the additional between-scan measurement error ($\sigma_\epsilon^2 = 0.0025$) of Frost’s Table II. We take observation interval $t = 1$ year and treatment effect $\Delta = 0.25$ (25% reduction in atrophy rate relative to placebo rate of approximately 1.0% brain volume loss per year). Using the combined residual, the conventional design reproduces Frost’s reported sample sizes (Table III). Using only the 0.0025 measurement-error term, as in earlier drafts, understates the residual by two orders of magnitude and yields degenerate sample sizes.

AD imaging trial sample sizes (per group):

Standard (r=0, k=2, f=0): 385

Run-in (r=2, k=2, f=0): 145 (62% reduction)

Full (r=2, k=2, f=2): 69 (82% reduction)

6 Discussion

The results presented here formalize and extend the framework of² by providing explicit variance formulas for the treatment effect estimator under designs that incorporate run-in and common close observations. Three points warrant discussion.

When run-in observations are most beneficial. The efficiency gain from run-in observations depends on the ratio $\sigma_b^2 t^2 / \sigma^2$. When between-subject slope variability is large relative to residual variance, as in neuroimaging outcomes, even one or two run-in observations yield

substantial sample size reductions. On the other hand, for cognitive outcomes, where this ratio is typically less favorable, the gains are more modest, and the cost of extended study duration must be weighed against the sample size reduction.

Common close as a low-cost efficiency gain. The common close phase provides information about individual trajectories without requiring participants to remain on treatment. Since treatment has ceased, both groups contribute to estimation of the underlying slope variability, which effectively tightens the precision of the treatment effect estimator. As with run-in observations, the marginal benefit of each additional common close observation diminishes. Nevertheless, the first one or two common close visits typically provide meaningful sample size reductions, particularly when $\sigma_b^2 t^2 / \sigma^2$ is large.

Relationship to the two-period model. The formulas derived here are consistent with the two-period linear mixed effects model of³. Their Scenario 1 (common slope across periods) corresponds to setting $\delta = \beta$ in our model, while their Scenario 2 (different slopes) corresponds to the general case. The advantage of our formulation lies in the explicit separation of run-in, treatment, and common close phases, which enables direct optimization of the number of observations in each phase.

Limitations. The derivations assume equally spaced observations, linear trajectories, and a random intercept plus random slope covariance structure. Extensions to unequal spacing, nonlinear progression, and more complex random effects structures (e.g., random intercept-slope correlation) are possible within the same Woodbury identity framework, but they yield more complex expressions that may not admit simple closed forms. The general (J_0, J_1, J_2) result, and the $J_0 = 2$ case, are established via the package's numerical Woodbury routine, validated against a direct GLS computation and a Monte Carlo benchmark, rather than via an independent symbolic derivation. The Mathematica symbolic trail in this repository (`analysis/scripts/woodbury_variance.m`) covers only $J_0 \in \{0, 1\}$. Extending the symbolic derivation to $J_0 = 2$ and the general case remains future work and is not required for the numerical results reported here, which do not depend on the symbolic route. The formulas also assume no dropout. In practice, the Dawson-Lagakos pattern-mixture approach² can be applied to adjust for anticipated attrition.

7 Conflict of interest

The author declares no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The `runinpower` R package, the four companion manuscripts of this compendium, the Mathematica symbolic-derivation scripts, and the shared bibliography are openly available in the

511 project repository at <https://github.com/rgt47/runinpower>. We list the specific paths in the
512 Reproducibility section below.

513 10 Reproducibility

514 This manuscript was rendered with R 4.4.x. Principal packages used are the in-package
515 `runinpower` source (nine source files: `allocation`, `ancova`, `averaged`, `covariance`, `covariance-`
516 `ar1`, `design`, `power`, `replicated`, `variance`), `tinytest` (testing harness), and `rmarkdown`
517 plus `bookdown` for the manuscript build. Mathematica symbolic derivations are at
518 `analysis/scripts/` (`woodbury_variance.m`). This file covers exactly the $J_0 = 0$ (no run-in)
519 and $J_0 = 1$ (one run-in observation) cases; it does not extend to $J_0 = 2$ or the general
520 (J_0, J_1, J_2) case reported in this manuscript's General Formula sections. Those results are
521 established instead via the package's numerical Woodbury routine (`var_gamma_matrix()`,
522 see `R/variance.R`) and validated against a direct GLS computation and a Monte Carlo
523 benchmark in the Numerical Examples section below, not via an independent hand (symbolic)
524 derivation beyond $J_0 = 1$. A symbolic derivation trail for $J_0 = 2$ and the general case does
525 not currently exist in this repository; see the Limitations paragraph below.

526 **Random-number generator.** The variance formulas are deterministic given fixed input
527 parameters, so no RNG is exercised in their evaluation. The validation chunk in the Numerical
528 Examples section first compares the Woodbury routine with a direct GLS computation, in
529 which Σ is assembled and inverted without the Woodbury identity, over a grid of designs
530 that includes common-close configurations ($J_2 > 0$). It then runs a Monte Carlo benchmark
531 at $(J_0, J_1, J_2) = (2, 2, 2)$ with 2000 replicates of $m = 200$ participants per group, setting
532 `set.seed(20260418)` at its entry point and comparing the empirical variance of $\hat{\gamma}$ with the
533 analytic value.

534 **Data and code availability.** The R package source is at `R/`. Mathematica scripts are at
535 `analysis/scripts/`. The companion manuscripts are at `analysis/report/{02,03,04}-*/`.
536 The manuscript source and shared bibliography (`../references.bib`) are at `analysis/report/01-runin-powe`

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